Decimal Place Value

Name _____ Date _____ No calculator Version 1

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KEY WORDS decimal *(tau ā-ira)* · place value *(uara tū)* · tenths *(hautekau)* · hundredths *(haurau)*

A Tenths, hundredths and thousandths ●○○

Remember: The first place after the decimal point is tenths, the second is hundredths and the third is thousandths. In 3.47 the 4 is worth 4 tenths, or 0.4.

1. Write the value of the underlined digit.

a) $3.\underline{4}7$ _____ **b)** $0.8\underline{5}$ _____ **c)** $12.\underline{6}$ _____

d) $4.27\underline{5}$ _____ **e)** $0.30\underline{9}$ _____ **f)** $2\underline{5}.18$ _____

g) $6.0\underline{4}$ _____ **h)** $0.\underline{5}12$ _____ **i)** $7.9\underline{3}6$ _____

B Decimals as fractions ●●○

Remember: One decimal place means tenths, two mean hundredths and three mean thousandths: $0.7 = \frac{7}{10}$, $0.23 = \frac{23}{100}$ and $0.009 = \frac{9}{1{,}000}$.

2. Write each decimal as a fraction.

a) 0.7 = _____ **b)** 0.3 = _____ **c)** 0.23 = _____ **d)** 0.09 = _____

e) 0.61 = _____ **f)** 0.009 = _____ **g)** 0.127 = _____ **h)** 0.045 = _____

3. Write each fraction as a decimal.

a) $\frac{3}{10}$ = _____ **b)** $\frac{9}{10}$ = _____ **c)** $\frac{47}{100}$ = _____ **d)** $\frac{7}{100}$ = _____

e) $\frac{81}{100}$ = _____ **f)** $\frac{3}{1{,}000}$ = _____ **g)** $\frac{259}{1{,}000}$ = _____ **h)** $\frac{31}{1{,}000}$ = _____

C Comparing and ordering decimals ●●○

⚠ **Watch out:** A longer decimal is not always larger: 0.5 is greater than 0.45, because 0.5 is the same as 0.50.

4. Write <, > or = in each box.

a) $0.45 \square 0.5$ **b)** $3.2 \square 3.18$ **c)** $0.7 \square 0.70$ **d)** $1.09 \square 1.1$

e) $0.305 \square 0.35$ **f)** $12.4 \square 12.39$ **g)** $0.099 \square 0.1$ **h)** $6.25 \square 6.250$

5. Write each list in order, from smallest to largest.

a) 0.5, 0.45, 0.54, 0.405 _____ **b)** 3.2, 3.02, 3.22, 2.3 _____

c) 0.09, 0.9, 0.1, 0.19 _____ **d)** 7.5, 7.05, 7.55, 7.505 _____

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D Multiplying and dividing by 10, 100 and 1,000 ●●●

⚠ **Watch out:** Do not just "add a zero". Every digit moves: one place for 10, two places for 100 and three places for 1,000. So $3.7 \times 10 = 37$, not 3.70.

6. Work out.

a) 45×100 = _____ **b)** 3.7×10 = _____ **c)** $0.62 \times 1{,}000$ = _____ **d)** 208×10 = _____

e) $4{,}800 \div 100$ = _____ **f)** $56 \div 10$ = _____ **g)** $7 \div 100$ = _____ **h)** $930 \div 1{,}000$ = _____

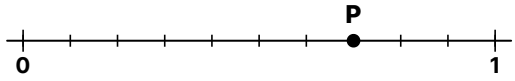
i) 1.05×100 = _____ **j)** 0.4×10 = _____ **k)** $2.5 \div 10$ = _____ **l)** $69 \div 100$ = _____

E Decimals on a number line ●●●

💡 **Remember:** First work out what each small step on the line is worth.

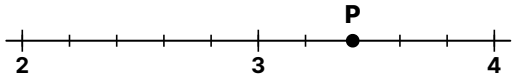
7. Write the number that the point P shows.

a)



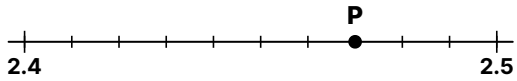
$P =$ _____

b)



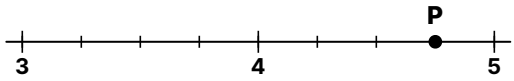
$P =$ _____

c)



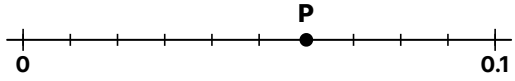
$P =$ _____

d)



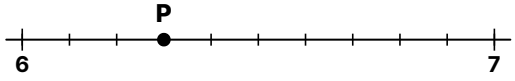
$P =$ _____

e)



$P =$ _____

f)



$P =$ _____

★ Extension

8. Write a number that is between each pair of numbers.

a) 0.3 and 0.4 **b)** 2.5 and 2.6 **c)** 0.07 and 0.08

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Decimal Place Value

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS decimal *(tau ā-ira)* · place value *(uara tū)* · tenths *(hautekau)* · hundredths *(haurau)*

A Tenths, hundredths and thousandths ●○○

Remember: The first place after the decimal point is tenths, the second is hundredths and the third is thousandths. In 3.47 the 4 is worth 4 tenths, or 0.4.

1. Write the value of the underlined digit.

a) $5.\underline{2}9$ _____ **b)** $0.6\underline{3}$ _____ **c)** $14.\underline{8}$ _____

d) $2.46\underline{1}$ _____ **e)** $0.70\underline{5}$ _____ **f)** $3\underline{8}.24$ _____

g) $9.0\underline{7}$ _____ **h)** $0.\underline{3}84$ _____ **i)** $1.6\underline{5}2$ _____

B Decimals as fractions ●●○

Remember: One decimal place means tenths, two mean hundredths and three mean thousandths: $0.7 = \frac{7}{10}$, $0.23 = \frac{23}{100}$ and $0.009 = \frac{9}{1{,}000}$.

2. Write each decimal as a fraction.

a) 0.9 = _____ **b)** 0.1 = _____ **c)** 0.37 = _____ **d)** 0.03 = _____

e) 0.59 = _____ **f)** 0.007 = _____ **g)** 0.213 = _____ **h)** 0.061 = _____

3. Write each fraction as a decimal.

a) $\frac{7}{10}$ = _____ **b)** $\frac{5}{10}$ = _____ **c)** $\frac{29}{100}$ = _____ **d)** $\frac{1}{100}$ = _____

e) $\frac{73}{100}$ = _____ **f)** $\frac{9}{1{,}000}$ = _____ **g)** $\frac{467}{1{,}000}$ = _____ **h)** $\frac{53}{1{,}000}$ = _____

C Comparing and ordering decimals ●●○

⚠ **Watch out:** A longer decimal is not always larger: 0.5 is greater than 0.45, because 0.5 is the same as 0.50.

4. Write <, > or = in each box.

a) $0.6 \square 0.58$ **b)** $4.17 \square 4.2$ **c)** $0.90 \square 0.9$ **d)** $2.1 \square 2.08$

e) $0.46 \square 0.406$ **f)** $15.29 \square 15.3$ **g)** $0.2 \square 0.199$ **h)** $3.40 \square 3.4$

5. Write each list in order, from smallest to largest.

a) 0.6, 0.56, 0.65, 0.506 _____ **b)** 4.1, 4.01, 4.11, 1.4 _____

c) 0.08, 0.8, 0.2, 0.18 _____ **d)** 2.5, 2.05, 2.55, 2.505 _____

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D Multiplying and dividing by 10, 100 and 1,000 ●●●

⚠ **Watch out:** Do not just "add a zero". Every digit moves: one place for 10, two places for 100 and three places for 1,000. So $3.7 \times 10 = 37$, not 3.70.

6. Work out.

a) 63×100 = _____ **b)** 5.2×10 = _____ **c)** $0.48 \times 1{,}000$ = _____ **d)** 307×10 = _____

e) $6{,}500 \div 100$ = _____ **f)** $84 \div 10$ = _____ **g)** $9 \div 100$ = _____ **h)** $710 \div 1{,}000$ = _____

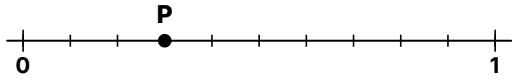
i) 2.06×100 = _____ **j)** 0.7×10 = _____ **k)** $3.5 \div 10$ = _____ **l)** $48 \div 100$ = _____

E Decimals on a number line ●●●

💡 **Remember:** First work out what each small step on the line is worth.

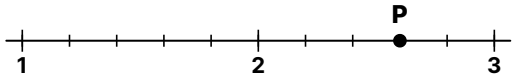
7. Write the number that the point P shows.

a)



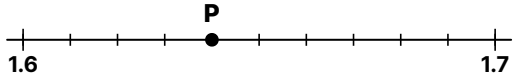
$P =$ _____

b)



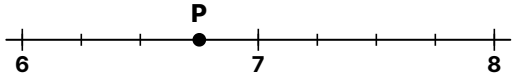
$P =$ _____

c)



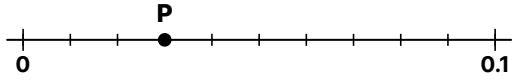
$P =$ _____

d)



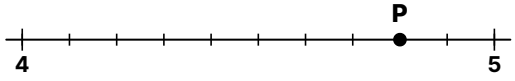
$P =$ _____

e)



$P =$ _____

f)



$P =$ _____

★ Extension

8. Write a number that is between each pair of numbers.

a) 0.6 and 0.7 **b)** 3.1 and 3.2 **c)** 0.04 and 0.05

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